

Contact

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Top Skills

Appium

Playwright

Java

Muhammad Tayyab

Senior SQA Engineer (8+ yrs) | Building Playwright + TypeScript Automation Frameworks | QA across Blockchain, AI, and Healthcare IT

Rawalpindi, Punjab, Pakistan

Summary

Senior SQA Engineer with 8+ years of experience designing and executing quality assurance strategies across the full SDLC specializing in manual, automation, API, and performance testing. I've spent my career in high-complexity, high-stakes domains where a missed defect isn't just inconvenient it's a compliance failure or a lost transaction: Blockchain/Web3 architecting Playwright + TypeScript automation frameworks, validating high-frequency APIs and distributed ledger data consistency Artificial Intelligence building adversarial testing frameworks to catch model bias and validate ML/NLP pipeline integrity Healthcare IT directing HIPAA-compliant EHR testing across medical billing and patient scheduling systems Recently, I've been deepening my automation skillset with Playwright + TypeScript, building Page Object Model frameworks with full CI/CD integration via GitHub Actions including cross-repository test triggering so my test suite runs automatically whenever the application under test changes. Tools & tech: Playwright, Selenium WebDriver, Cypress, Python, TypeScript, Java, Postman, RestAssured, JMeter, SQL/PostgreSQL, Git, GitHub Actions, JIRA. Always open to connecting with fellow QA engineers, or discussing SDET/automation roles. Portfolio: [<https://tayyabmirza1606.github.io/portfolio-v1/>] GitHub: [<https://github.com/Tayyabmirza1606>]

Experience

Codora

2 years 4 months

Senior SQA Manual and Automation Engineer

May 2024 - Present (2 years 4 months)

Islāmābād, Pakistan

Architected and maintained end-to-end automation frameworks using Python and TypeScript with Playwright, improving regression coverage for Web3

platforms by 50%. Executed rigorous API Testing using Postman and Swagger to validate secure, high-frequency RESTful endpoints serving blockchain data. Managed PostgreSQL database validation, auditing transaction consistency and real-time data persistence in distributed ledgers. Conducted comprehensive performance benchmarking to ensure platform stability during high-volume crypto trading events. Streamlined quality delivery by integrating automated suites into CI/CD pipelines, reducing production defect leakage through "shift-left" testing. Collaborated with decentralized protocol developers to ensure smart contract interactions align with security standards.

Senior Quality Assurance Automation Engineer

May 2024 - Present (2 years 4 months)

Architected and maintained end-to-end automation frameworks using Python and TypeScript with Playwright, improving regression coverage for Web3 platforms by 50%. Executed rigorous API Testing using Postman and Swagger to validate secure, high-frequency RESTful endpoints serving blockchain data. Managed PostgreSQL database validation, auditing transaction consistency and real-time data persistence in distributed ledgers. Conducted comprehensive performance benchmarking to ensure platform stability during high-volume crypto trading events. Streamlined quality delivery by integrating automated suites into CI/CD pipelines, reducing production defect leakage through 'shift-left' testing. Collaborated with decentralized protocol developers to ensure smart contract interactions align with security standards.

STech.ai

1 year 1 month

Lead SQA Engineer

June 2023 - June 2024 (1 year 1 month)

Islamabad, Islāmābād, Pakistan

Led the QA efforts for AI-based projects, collaborating closely with data scientists, AI and machine learning engineers, and software developers to validate AI models, algorithms, and data pipelines.

Designed and implemented AI-specific testing methodologies, including data validation, model performance evaluation, and adversarial testing, to uncover potential biases, errors, and vulnerabilities in AI systems.

Software Quality Assurance Lead

June 2023 - May 2024 (1 year)

Spearheaded QA operations for AI-driven platforms, defining specialized validation protocols for ML models, data pipelines, and NLP algorithms.

Established Adversarial Testing frameworks to identify model biases and ensure the robustness of AI decision-making engines. Collaborated with Data Science teams to implement automated data validation checks, ensuring 99.9% data integrity for large-scale training sets. Mentored junior QA engineers on AI testing best practices and the automation of repetitive Computer Vision validation tasks. Managed the full defect lifecycle using Jira, providing analytical reports on model performance and system stability to executive stakeholders.

CareCloud

5 years 2 months

Senior Software Quality Assurance Engineer

December 2020 - June 2023 (2 years 7 months)

Directed full-lifecycle testing for enterprise-level Electronic Health Record (EHR) systems, ensuring strict HIPAA compliance and data privacy. Developed scalable automation frameworks using Selenium WebDriver with Java, focusing on core medical billing and patient scheduling modules. Executed high-volume Performance and Load testing using JMeter to verify system resilience during peak usage periods. Authored comprehensive Test Plans and Strategies for new product features, coordinating with Product Managers to align QA with business objectives. Utilized SQL Server for deep-dive backend validation, ensuring 100% data accuracy during migrations of sensitive patient records. Led defect triage meetings and facilitated cross-functional communication to resolve high-priority production blockers.

SQA Engineer

May 2018 - June 2023 (5 years 2 months)

Islamabad, Islāmābād, Pakistan

Performing manual and automation testing on products. Work closely on the deadlines of the project with team members. Write efficient and according to the requirements test cases and test scripts. Perform manual and automation testing on the products as per client requirements and meet client's satisfaction on changes. Perform load and stress testing of application and database servers. Identify, analyzes, and document defects, questionable functions, errors, and inconsistencies in software program functions, outputs, and content. Develop, execute and evaluate manual test cases and report test results, and keep track of all changes. Writing, review and execution of test scenarios manually. Working on end to end SDLC, agile testing methodologies and test phases including unit, integration, system, regression and performance testing. Create and execute different queries in SQL during

the testing of applications. Responsible to create detailed test cases and test scripts. Communicate effectively with development teams to get the issues resolved and addressed. Set expectations of the seniors on tasks and projects and then deliver on time. Reportable to QA Lead/hierarchical management and perform any task assigned by the seniors. Responds to inquiries from users and technical support regarding functions and operations of software program.

ITS Tech World

Associate Software Quality Assurance Engineer

March 2018 - September 2018 (7 months)

Gujranwala

Executed functional and UI testing for diverse web applications, identifying navigation and checkout workflow defects. Authored initial test scenarios and documentation, establishing baseline QA processes for development teams.

Education

PMAS Arid Agriculture University, Rawalpindi

Bachelor of Science, Computational Science · (February 2013 - September 2017)